Chapter 4: Descriptive Statistics in Action

Understanding Your Data Through Numbers

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0.1 Overview

This reading supports the week 12 lab session where you'll explore descriptive statistics using the DataLab data. The goal is to understand what these statistics tell us about our data.

By the end of this reading, you'll be able to:

- Interpret means, medians, and standard deviations
- Understand what the range and quartiles tell us
- Recognise when different measures are appropriate
- Use R to compute these statistics (with help)

0.2 Part 1: Describing the Centre

When we collect data, the first question is usually: **What's typical?**

0.2.a The Mean: The Balancing Point

The mean is the arithmetic average. If we measured 5 people's memory scores:

```
{webr-r}
# Five memory scores
memory_scores <- c(4, 6, 7, 8, 10)

# Calculate the mean
mean(memory_scores)
```

The mean is 7. This is the "balancing point" — if you put these scores on a seesaw, 7 is where it would balance.

0.2.b The Median: The Middle Value

The median is the middle value when data are sorted. Half the values are above it, half below.

```
{webr-r}
# Same five scores
memory_scores <- c(4, 6, 7, 8, 10)
```

```
# The median
median(memory_scores)
```

With 5 values, the median is the 3rd value (when sorted). Here, both mean and median are 7.

0.2.c When Mean and Median Differ

Sometimes they tell different stories. Consider these scores:

```
{webr-r}
# One person did exceptionally well
memory_with_outlier <- c(4, 6, 7, 8, 25)

# Compare mean and median
mean(memory_with_outlier)
median(memory_with_outlier)
```

The mean shoots up to 10, but the median stays at 7. The median is **resistant** to extreme values (outliers). The mean is pulled by them.

When to use which?

- Mean: When data are roughly symmetric, no extreme outliers
- Median: When data are skewed or have outliers (like income data)

0.3 Part 2: Describing the Spread

The centre isn't the whole story. Two groups can have the same mean but look completely different:

```
{webr-r}
# Consistent group
group_a <- c(6, 7, 7, 7, 8)

# Variable group
group_b <- c(2, 4, 7, 10, 12)

# Same mean!
mean(group_a)
mean(group_b)
```

Both means are 7, but the groups are very different. We need measures of **spread**.

0.3.a The Range: Simplest Spread

The range is just the distance from lowest to highest:

```
{webr-r}
# Range gives you min and max
range(group_a)
range(group_b)

# Calculate the spread
max(group_a) - min(group_a)
max(group_b) - min(group_b)
```

Group A spans 2 points (6 to 8). Group B spans 10 points (2 to 12). The range immediately shows Group B is more variable.

Limitation: The range only uses two values, so it's sensitive to outliers.

0.3.b The Standard Deviation: The Gold Standard

The **standard deviation (SD)** measures how far scores typically fall from the mean.

```
{webr-r}
# SD for each group
sd(group_a)
sd(group_b)
```

Group A's SD is about 0.7 — scores cluster tightly around 7. Group B's SD is about 4.2 — scores are much more spread out.

Rule of thumb: In roughly normal data, about 68% of values fall within 1 SD of the mean.

If the mean is 7 and SD is 1:

- ~68% of scores fall between 6 and 8

0.3.c The summary() Function: Everything at Once

R's `summary()` function gives you a quick overview:

```
{webr-r}
# Complete summary
summary(group_b)
```

This shows:

- Min and Max (the range endpoints)
- 1st and 3rd Quartile (the middle 50%)
- Median (the middle value)
- Mean

0.4 Part 3: Seeing Descriptives in Psychology Data

Let's look at what descriptive statistics tell us with realistic psychology examples.

0.4.a Example 1: Reaction Times

Imagine we measured 10 participants' reaction times (in milliseconds):

```
{webr-r}
# Reaction times from 10 participants
reaction_times <- c(245, 267, 278, 298, 312, 289, 301, 256, 334, 271)

# Central tendency
mean(reaction_times)
median(reaction_times)

# Spread
sd(reaction_times)
range(reaction_times)
```

Interpretation:

- Mean reaction time is about 285ms
- The median (284ms) is similar, suggesting roughly symmetric data
- SD of about 28ms tells us most people fall within ±28ms of the mean
- Range of 245 to 334ms shows the fastest and slowest participants

0.4.b Example 2: Memory Scores by Condition

Now let's compare two groups. Lab A saw abstract words, Lab B saw concrete words:

```
{webr-r}
# Lab A: Abstract words
lab_a <- c(4, 5, 6, 5, 7, 4, 6, 5, 5, 6)

# Lab B: Concrete words
lab_b <- c(6, 7, 8, 7, 9, 6, 8, 7, 8, 7)

# Compare means
mean(lab_a)
mean(lab_b)

# Compare SDs
sd(lab_a)
sd(lab_b)
```

Interpretation:

- Lab A mean: ~5.3 words recalled
- Lab B mean: ~7.3 words recalled
- The difference is 2 words — but is that meaningful?
- Both SDs are around 1, so 2 words is about **2 standard deviations** — a substantial difference

0.4.c Example 3: Pre-Post Mood Ratings

Same people rated their mood before and after the session:

```
{webr-r}
# Before session (1-10 scale)
mood_pre <- c(5, 6, 4, 7, 5, 6, 4, 5, 6, 5)

# After session
mood_post <- c(6, 7, 6, 8, 6, 7, 5, 6, 7, 6)

# Means
mean(mood_pre)
mean(mood_post)

# The change for each person
mood_change <- mood_post - mood_pre
mean(mood_change)
sd(mood_change)
```

Interpretation:

- Mean mood increased by 1.1 points on average
- The SD of change (0.7) tells us most people improved by 0.4 to 1.8 points
- This is a within-subjects comparison — same people before and after

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0.5 Part 4: What the Numbers Mean

0.5.a Standard Deviation in Context

The SD only makes sense relative to the mean and the scale.

```
{webr-r}
# Memory (0-10 scale)
memory <- c(5, 6, 7, 5, 6, 7, 6, 5, 6, 7)
mean(memory)
sd(memory)

# Reaction time (milliseconds)
rt <- c(250, 280, 310, 240, 290, 300, 260, 270, 290, 280)
mean(rt)
sd(rt)
```

- Memory: Mean = 6, SD = 0.77. An SD of 0.77 on a 10-point scale is fairly consistent.
- RT: Mean = 277ms, SD = 23ms. Is that a lot of variability? It's about 8% of the mean — fairly typical for reaction times.

0.5.b When Means Can Mislead

Consider these two possible DataLab results:

```
{webr-r}
# Scenario A: Everyone similar
scenario_a <- c(5, 5, 6, 6, 6, 6, 7, 7)

# Scenario B: Bimodal (two clusters)
scenario_b <- c(2, 2, 3, 3, 9, 9, 10, 10)

# Same mean!
mean(scenario_a)
mean(scenario_b)

# But very different SDs
sd(scenario_a)
sd(scenario_b)
```

Both means are 6, but Scenario B's huge SD (3.7) signals that something interesting is happening — perhaps there are two distinct groups.

0.6 Key Terms

Term	Definition
Mean	Arithmetic average (sum ÷ count)
Median	Middle value when sorted
Standard Deviation	Average distance from the mean
Range	Difference between max and min
Quartiles	Values that divide data into four equal parts
Outlier	A value far from the others

0.7 Summary

Question	Statistic to Use
What's typical?	Mean or Median
How spread out?	SD or Range
Any extreme values?	Compare Mean vs Median
Quick overview?	`summary()`
Compare groups?	Mean and SD for each

0.8 Additional Resources

If you want to explore further:

- R for Data Science — Free online textbook
- Quick-R Statistics — Syntax reference